Compiled Solutions – Senior Division

1. **00-01 C4 Prefix/Infix/Postfix**

. The equation a 2 ↑ b 2 ↑- a b + a b - * = translates to infix equation $a^2 - b^2 = (a + b)(a - b)$ which translates to prefix equation = – ↑ a 2 ↑ b 2 * + a b – a b

= – ↑ a 2 ↑ b 2 * + a b – a b

2. **00-01 C4 Prefix/Infix/Postfix**

The expression translates to
((1 # 5) + (2 # 8)) % 2 = (3 + 5) % 2 = $16^{1/2} = 4$

4

3. **01-02 C4 Prefix/Infix/Postfix**

The postfix expression translates to infix as:
A + B / (C * (D + E - A)) + F - E and substituting
7 + 9 /(3 * (4 + 6 - 7)) + 8 - 6 = 7 + 1 + 8 - 6 = 10

10

4. **01-02 C4 Prefix/Infix/Postfix**

The prefix expression translates to infix as: (X + 4) * (6 - X)
and setting this equal to 0 gives integer solutions of -4 and 6.

-4 and 6

5. **02-03 C4 Prefix/Infix/Postfix**

7 3 + 2 6 * 2 3 ↑ 3 + @ = (7 + 3) (2 * 6) (2 ↑ 3 +3) @ =
@ (10 12 11) = 10

10

6. **02-03 C4 Prefix/Infix/Postfix**

/ - + ↑ X 2 * * 2 X Y ↑ Y 2 – X Y translates to
$(X^2 + 2XY - Y^2) / (X - Y)$ in infix.
This translates to X 2 ↑ 2 X * Y * + Y 2 ↑ - X Y - / in postfix.

As shown

7. **03-04 C4 Prefix/Infix/Postfix**

$$\frac{(A^2 + B)^2 - (AB + B^2)^{\frac{1}{2}}}{A / B^2 + A^2 B / C}$$ translates to the following postfix expression:

A 2 ↑ B + 2 ↑ A B * B 2 ↑ + 1 2 / ↑ - A B 2 ↑ / A 2 ↑ B * C / + /

As shown

8. **03-04 C4 Prefix/Infix/Postfix**

@ @ # 2 4 # 4 4 * 4 # 1 5 = @ @ (2 # 4) (4 # 4) * 4 (1 # 5) = 13
@ @ 3 4 * 4 3 = @ (3 @ 4) (4 * 3) = 5 @ 12 = 13

9. **04-05 C4 Prefix/Infix/Postfix**

+ * # 3 4 # 6 2 * ↑ 2 3 3 / * 4 5 + 2 3 1 = 12
* + * $\sqrt{12}$ $\sqrt{12}$ * (8) 3 / (20) (5) 1 = # + $\sqrt{144}$ 24 4 1=
* + 12 24 4 1 = # * 36 4 1 = # 144 1 = $\sqrt{144}$ = 12

10. **04-05 C4 Prefix/Infix/Postfix**

A B C + / B 2 ↑ C B / + - 2 ↑ converts to infix as: As shown
$\left[\frac{A}{B+C}-\left(B^2+\frac{C}{B}\right)\right]^2$ which converts to prefix as:
↑ - / A + B C + ↑ B 2 / C B 2

11. **05-06 C4 Prefix/Infix/Postfix**

$AB * CD - / 2 \uparrow B3 \uparrow C2 \uparrow / + CD2 \uparrow * B12 / \uparrow / -$ translates to: -11
$\left(\frac{AB}{C-D}\right)^2+\frac{B^3}{C^2}-\frac{CD^2}{\sqrt{B}}$ and evaluates to 4 + 1 - 16 = -11

12. **05-06 C4 Prefix/Infix/Postfix**

The prefix expression $\uparrow - + / AB / + AB - AB / * AB / AB2$ As shown
becomes the following postfix expression:
$AB / AB + AB - / + AB * AB // - 2 \uparrow$

13. **06-07 C4 Prefix/Infix/Postfix**

/ ↑ / + ↑A 2 ↑ - B C 2 - * A B ↑ C 2 / 1 2 ↑ / ↑ - A * B C 4 – B C / 1 3 As shown

14. **06-07 C4 Prefix/Infix/Postfix**

$(A^2 + B)^2 – (B^2 + A) * (A^2 + A^2 + B) =$ -12
$(2^2 - 1)^2 – ((-1)^2 + 2) * (2^2 + 2^2 - 1) =$
$3^2 - 3 * 7 = 9 – 21 = -12$

15. **07-08 C4 Prefix/Infix/Postfix**

prefix: / * + A B – B C ↑ A C — As shown
infix: ((A + B) * (B + C)) / (A ↑ C)
postfix: A B + B C - * A C ↑ /

16. **07-08 C4 Prefix/Infix/Postfix**

4 3 * 9 7 + % 4 5 + 2 ↑ 3 5 * @ / — 16
= (4 * 3) (9 + 7) % (4 + 5) 2 ↑(3 *5) @ /
= (12 % 16) (9 ↑ 2) 15 @ / = 48 (81 @ 15) /
= 48 / 3 = 16

17. **08-09 C4 Prefix/Infix/Postfix**

$(A^2 + B) * C / B - (B + C) / A + A^3 * B^2$ converts to the following prefix expression: + - / * + ↑ A 2 B C B / + B C A * ↑ A 3 ↑ B 2 — As shown

18. **08-09 C4 Prefix/Infix/Postfix**

4 3 6 * 9 3 + @ 3 2 ↑ @ * 2 3 5 * * 7 5 * @ % — 2
= (4 (((3 * 6) @ (9 + 3)) @ 3^2)) % ((2 * 3 * 5) @ (7 * 5))
= (4 (18 @ 12)) % (30 @ 35)
= (4 * (6 @ 9)) % 5 = (4 * 3) % 5 = 12 % 5 = 2

19. **09-10 C4 Prefix/Infix/Postfix**

prefix: + / A B / * ↑ C 2 / + A B C / + * A B C ↑ B 2 — As shown
infix: (A / B) + (C^2 * ((A + B) / C)) / ((A B + C) / B^2)
postfix: A B / C 2 ↑ A B + C / * A B * C + B 2 ↑ / / +

20. **09-10 C4 Prefix/Infix/Postfix**

1 2 3 * + 3 2 ↑ + 2 2 ↑ 8 + @ 5 2 * 6 3 / - % — 0
=1 (2 * 3) + 3^2 + 2^2 8 + @ (5 * 2) (6 / 3) - %
= (1 + 6) 9 + (4 + 8) @ (10 - 2) %
= (7 + 9) 12 @ 8 % = (16 @ 12) 8 % = 4 % 8 = 0

21. **10-11 C4 Prefix/Infix/Postfix**

Postfix: C A B ↑ B A ↑ + A B * ↑ * A B * C + B ↑ / — As shown
Prefix: / * C ↑ + ↑ A B ↑ B A * A B ↑ + * A B C B

22. **10-11 C4 Prefix/Infix/Postfix**

4 3 6 * 9 3 + @ 3 2 ↑ @ * 2 3 5 * * 7 5 * @ % = 2
(4 (((3 * 6) @ (9 + 3)) @ 3^2)) % ((2 * 3 * 5) @ (7 * 5))
= (4 (18 @ 12)) % (30 @ 35)
= (4 * (6 @ 9)) % 5 = (4 * 3) % 5 = 12 % 5 = 2

23. **11-12 C2 Prefix/Infix/Postfix**

2 4 6 + * 5 / 3 2 ↑ 4 2 – * + = 2 (4 6 +) * 5 / (3 2 ↑) (4 2 –) * + 22
= (2 10 *) 5 / (9 2 *) + = (20 5 /) 18 + = 4 18 + = 22

24. **11-12 C2 Prefix/Infix/Postfix**

Prefix: + – * A ↑ B C / B A * C + / A B C As shown

Infix: A * B^C – B/A + C * (A/B + C)
= A * (B C ↑) – (B A /) + C * (A B / C +)
= A B C ↑ * B A / – C A B / C + * +

25. **12-13 C2 Prefix/Infix/Postfix**

A B + C / B E ↑ * D A + 2 ↑ + B C E * + A / – 8
= (A B +) C / (B E ↑) * (D A + 2 ↑)+ B (C E *) + A / –
= (A + B) / C * B^E + $(D + A)^2$ – (B + C * E) / A
= (2 + 4) / 6* 4^0 + $(1 + 2)^2$ – (4 + 6 *0) / 2
= 6 / 6 * 1 + 3^2 – 4 / 2 = 1 * 1 + 9 – 2 = 1 + 9 – 2 = 8

26. **12-13 C2 Prefix/Infix/Postfix**

– + – * + A B –↑ A 2 B / + A B ↑ C 2 / A B C As shown
= – + – * (+ A B) – (↑ A 2) B / (+ A B)(↑ C 2)(/ A B) C
= (A + B) * (A^2 – B) – (A + B) / C^2 + A / B – C INFIX
= (A B +) * (A 2 ↑ B –) – (A B + C 2 ↑ /) + (A B /) C
= (A B + A 2 ↑ B – *) – (A B + C 2 ↑ /) + (A B /) C
= A B + A 2 ↑ B – * A B + C 2 ↑ / – A B / + C –

27. **13-14 C2 Prefix/Infix/Postfix**

– + A B * / * C ↑ D 2 A B = -24
– (A + B) * / * C (D^2) A B = – (A + B) * / (C * D^2) A B =
– (A + B) * (C * D^2 / A) B = – (A + B) (C * D^2 / A * B) =
A + B – C * D^2 / A * B = 8 + 4 – 2 * 6^2 / 8 * 4 =
8 + 4 – 2 * 36 / 8 * 4 = 8 + 4 – 36 = 12 – 36 = -24

28. **13-14 C2 Prefix/Infix/Postfix**

$\frac{A(B+C)}{A-C}+\frac{A+B}{C^2}$

= [A (B + C)] / (A - C) + (A + B) / C^2
= [A B C + *] / [A C -] + [A B +] / [C 2 ↑]
= [A B C + * A C - /] + [A B + C 2 ↑ /]
= A B C + * A C - / A B + C 2 ↑ / +

29. **14-15 C2 Prefix/Infix/Postfix**

A + B * (C + A^2) – B / (C + A) = A + B * (C A 2 ↑ +) – B / (C A +) As shown
= A + (B C A 2 ↑ + *) – (B C A + /)
= (A B C A 2 ↑ + * +) – (B C A + /)
= A B C A 2 ↑ + * + B C A + / –

30. **14-15 C2 Prefix/Infix/Postfix**

– # * 8 6 / 4 2 – @ * @ 9 4 3 * ↑ 5 2 2 ↑ # 8 2 3 120
= – # (* 8 6)(/ 4 2) – @ * (@ 9 4) 3 * (↑ 5 2) 2 ↑ (# 8 2) 3
= – (# 48 2) – @ (* 6 3) (* 25 2) (↑ 5 3)
= – 25 – (@ 18 50) 125 = – 25 (– 30 125) = – 25 -95 = 120

31. **15-16 C2 Prefix/Infix/Postfix**

Inserting symbols and parentheses, the expression becomes: As shown
((A + B) * (C – A) * (C / B – A)) / (A * B – C)
= ((+ A B) * (– C A) * (– / C B A)) / (– * A B C)
= (* * + A B – C A – / C B A) / (– * A B C)
= / * * + AB – C A – / C B A – * A B C

32. **15-16 C2 Prefix/Infix/Postfix**

6 4 # 9 4 % + 3 1 # ↑ 7 9 + 3 9 * 2 – % 5 / % 22
= (6 4 #)(9 4 %) + (3 1 #) ↑ (7 9 +)(3 9 *) 2 – % 5 / %
= (5 6 +) 2 ↑ 16 (27 2 –) % 5 / %
= (11 2 ↑) (16 25 %) 5 / %
= 121 (20 5 /) % = 121 4 % = 22